

Phase 3

Project Design Phase

Project: **Implement Client Script & UI Policy (Incident)**

Item	Details
Platform	ServiceNow (Incident Management)
Organization	SmartBridge
Team ID	SWTID-2026-3598
Team Size	5
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1. Design Overview

The solution is designed as a layered set of client-side controls on the Incident table. The **Impact** field is the single driver: its value decides whether the UI Policy conditions evaluate to true, triggers the onChange script and is checked again by the onSubmit script at save time. The onCellEdit script is independent and protects the State field in list views.

2. Solution Architecture

Layer	Component	Table / Field	Trigger
Declarative	UI Policy: High Impact Control	incident; condition Impact is 1 - High	On load and on change of form
Declarative	UI Policy Action 1	assignment_group - Mandatory True, Read-only False	UI Policy condition true
Declarative	UI Policy Action 2	urgency - Read-only True, Visible Leave alone	UI Policy condition true
Scripted	Client Script: Auto set urgency for high impact	incident; onChange; field Impact	Impact value changes
Scripted	Client Script: Prevent save if Assigned To missing	incident; onSubmit	User clicks Submit / Update
Scripted	Client Script: Prevent state change via list edit	incident; onCellEdit; field State	Cell edited in list

3. Logical Flow (Form Interaction)

```
User opens Incident form
|
v
User sets Impact = 1 - High
|
+--> UI Policy condition TRUE
|     - Assignment group = Mandatory
|     - Urgency = Read-only
|
+--> onChange script fires
|     - Urgency set to 1 - High
|     - Info message shown
v
User clicks Submit / Update
|
v
onSubmit script: Impact = High AND Assigned To empty ?
|-- YES --> error box on Assigned To, save cancelled
|-- NO  --> record saved

If Impact changes away from High:
    UI Policy 'Reverse if false' undoes mandatory / read-only
```

4. Detailed Component Design

4.1 UI Policy - High Impact Control

Property	Value
Table	Incident [incident]
Short description / Name	High Impact Control
Application	Global
Active	true
Order	100
Condition	Impact is 1 - High
Global	true (applies to all form views)
On load	true
Reverse if false	true
Inherit	false

4.2 UI Policy Actions

Field	Mandatory	Visible	Read-only	Clear field value
Assignment group	True	Leave alone	False	unchecked
Urgency	Leave alone	Leave alone	True	unchecked

4.3 Client Script - onChange (Auto set urgency for high impact)

Type: onChange, Field: Impact, UI Type: Desktop, Active, Global, Isolate script enabled. The script returns immediately while the form is loading or when the new value is blank, so existing records are not modified on open. When the new value is '1' it sets Urgency to '1' and shows an information message.

```
function onChange(control, oldValue, newValue, isLoading) {
    if (isLoading || newValue == '') {
        return;
    }
    if (newValue == '1') {
        g_form.setValue('urgency', '1');
        g_form.addInfoMessage('Urgency set to High for High impact incident.');
    }
}
```

4.4 Client Script - onSubmit (Prevent save if Assigned To missing)

Type: onSubmit. Returning *false* cancels the submission; *showErrorBox* places a red message under the Assigned To field.

```
function onSubmit() {
    if (g_form.getValue('impact') == '1' &&
        g_form.getValue('assigned_to') == '') {
        g_form.showErrorBox(
            'assigned_to',
            'Assigned To is mandatory for High impact incidents.'
        );
    }
}
```

```
    return false;
  }
  return true;
}
```

4.5 Client Script - onCellEdit (Prevent state change via list edit)

Type: onCellEdit, Field: State. Calling *callback(false)* cancels the inline edit, so the value stays unchanged.

```
function onCellEdit(sysIDs, table, oldValues, newValue, callback) {
  alert('State cannot be updated using list editing. Please open the Incident.');
```

```
  callback(false);
}
```

5. Design Decisions and Rationale

- **UI Policy for mandatory/read-only:** declarative, no code and automatically reversible with 'Reverse if false'.
- **Client Script for auto-set and validation:** UI Policies cannot assign values conditionally with messages nor cancel a submit.
- **Separate scripts per event:** each script has one responsibility, which simplifies testing and maintenance.
- **Isolate script enabled:** follows ServiceNow strict-mode guidance (no direct DOM, jQuery or window access).

6. Risks and Design Limitations

- Client-side rules do not apply to REST/SOAP APIs or imports; a server-side Business Rule is recommended for full enforcement.
- The onCellEdit script blocks *all* list edits of State, for all roles; a role check could be added if exceptions are needed.
- The onSubmit script checks only the High impact condition, as specified by the business rule.